

Excel Formulas

Basic Functions

SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, COUNTBLANK

1. SUM()

SUM() adds numbers together.

It is one of the most frequently used Excel functions.

***Syntax**

=SUM(number1,number2,...)

Most commonly:

=SUM(range)

***Example**

=SUM(E2:E11)

This adds all employee salaries.

***What is the total salary expense of the company?**

=SUM(E2:E11)

2. AVERAGE()

AVERAGE() calculates the arithmetic mean.

***Syntax**

=AVERAGE(range)

***Example**

=AVERAGE(E2:E11)

***What is the average employee salary?**

=AVERAGE(E2:E11)

3. MIN()

MIN() returns the smallest numerical value.

*Syntax

=MIN(range)

*Example

=MIN(E2:E11)

*What is the lowest employee salary?

=MIN(E2:E11)

4. MAX()

MAX() returns the largest numerical value.

*Syntax

=MAX(range)

*Example

=MAX(E2:E11)

*What is the highest employee salary?

=MAX(E2:E11)

5. COUNT()

COUNT() counts cells containing **numbers**.

*Syntax

=COUNT(range)

*Example

=COUNT(E2:E11)

*Important

COUNT() does **not** count normal text.

6. COUNTA()

COUNTA() counts **non-empty cells**.

It counts:

- Numbers
- Text

- Dates
- Errors
- Other non-empty values

***Syntax**

=COUNTA(range)

***Example**

=COUNTA(B2:B11)

7. COUNTBLANK()

COUNTBLANK() counts empty cells.

***Syntax**

=COUNTBLANK(range)

***Example**

=COUNTBLANK(B2:B11)

***COUNT vs COUNTA vs COUNTBLANK**

Function	Purpose
COUNT()	Counts numbers
COUNTA()	Counts non-empty cells
COUNTBLANK()	Counts blank cells

Interview shortcut

Remember:

COUNT = Numbers

COUNTA = Anything filled

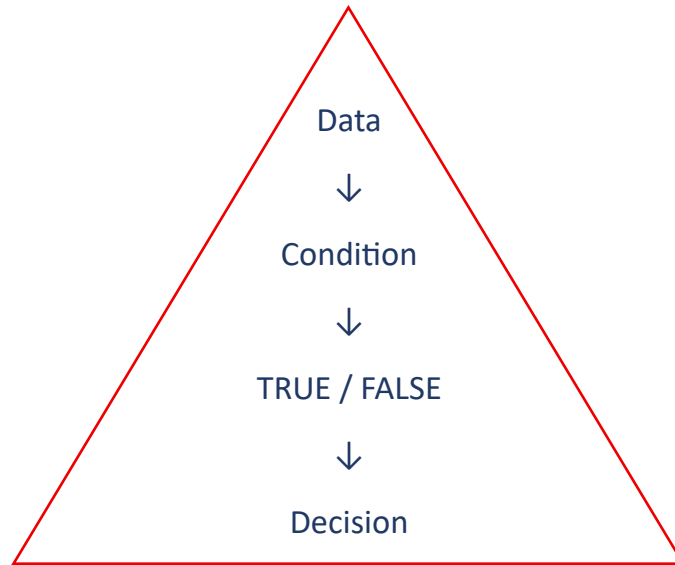
COUNTBLANK = Empty

Logical Functions

IF, IFS, AND, OR, NOT, IFERROR

Logical functions allow Excel to make decisions.

Think:



1. IF()

IF() checks a condition and returns one result if TRUE and another if FALSE.

***Syntax**

=IF(logical_test,value_if_true,value_if_false)

***Employee Example**

Business requirement:

Employees earning ₹50,000 or more should be classified as "High Salary"; otherwise "Standard Salary".

Add a new column:

Salary_Category

=IF(E2>=50000,"High Salary","Standard Salary")

*Result

Name	Salary	Salary Category
Rahul	45000	Standard Salary
Priya	38000	Standard Salary
Amit	52000	High Salary
Sneha	48000	Standard Salary
Rohan	42000	Standard Salary
Neha	60000	High Salary
Akash	55000	High Salary
Pooja	35000	Standard Salary
Karan	46000	Standard Salary
Anjali	58000	High Salary

*Real Business Uses of IF()

IF() is used for:

- Employee eligibility
- Bonus eligibility
- Customer classification
- Profitability
- Sales targets
- Pass/fail
- Risk classification
- Inventory status
- Payment status

2. IFS()

IFS() is useful when you have **multiple conditions**.

Instead of repeatedly nesting IF statements, you can test multiple conditions.

*Example

HR wants salary categories:

$\geq 60000 \rightarrow$ Very High

$\geq 50000 \rightarrow$ High

$\geq 40000 \rightarrow$ Medium

$< 40000 \rightarrow$ Low

***Formula:**

```
=IFS(  
E2>=60000,"Very High",  
E2>=50000,"High",  
E2>=40000,"Medium",  
E2<40000,"Low"  
)
```

 Important IFS Rule

Order matters.

Excel evaluates conditions from **left to right** and returns the result for the first TRUE condition.

3. AND()

AND() checks whether **all conditions are TRUE**.

***Syntax**

```
=AND(condition1,condition2,...)
```

***Example:**

```
=AND(E2>=50000,F2>=4)
```

Meaning:

Salary $\geq$ ₹50,000 **AND** Experience $\geq$ 4 years.

4. AND + IF

This is much more useful.

***Business rule:**

Employee receives a bonus if salary is at least ₹50,000 AND experience is at least 4 years.

***Formula:**

```
=IF(AND(E2>=50000,F2>=4),"Eligible","Not Eligible")
```

*Results

Employee	Salary	Experience	Bonus
Amit	52000	4	Eligible
Neha	60000	5	Eligible
Akash	55000	4	Eligible
Anjali	58000	4	Eligible

5. OR()

OR() returns TRUE when **at least one condition is TRUE**.

***Syntax**

=OR(condition1,condition2,...)

***Example:**

=OR(E2>=60000,F2>=5)

Meaning:

Salary $\geq$ ₹60,000 **OR** experience $\geq$ 5 years.

6. OR + IF

Business rule:

Give an employee a special review if they have either salary $\geq$ ₹60,000 OR experience $\geq$ 5 years.

=IF(OR(E2>=60000,F2>=5),"Special Review","Normal Review")

7. NOT()

NOT() reverses TRUE/FALSE.

***Syntax**

=NOT(logical)

***Example:**

=NOT(E2>=50000)

If: E2 = 45000

Then: E2>=50000 $\rightarrow$ FALSE

NOT reverses it: TRUE

Simple example

=NOT(TRUE)

Result: FALSE

=NOT(FALSE)

Result: TRUE

8. IFERROR()

IFERROR() lets you replace an Excel error with a useful result.

*Syntax

=IFERROR(value,value_if_error)

*Example

Suppose:

Sales	Orders
100000	100
50000	50
20000	0

You want:

Average Order Value = Sales / Orders

Formula:

=A2/B2

For the third row: 20000 / 0

Excel gives: #DIV/0!

Instead:

=IFERROR(A2/B2,"No Orders")

Now Excel displays:

No Orders

instead of an ugly error.

*Real Data Analyst Use

IFERROR() is extremely useful when:

- Dividing by zero
- Lookup doesn't find a value

- Data contains errors
- Building dashboards
- Cleaning reports
- Preparing management reports

DATASET — SALES PERFORMANCE

Salesperson	Region	Sales	Target	Experience	Customer_Rating
Asha	West	120000	100000	5	4.8
Rahul	North	85000	100000	3	4.2
Priya	South	150000	120000	6	4.9
Amit	West	95000	100000	4	3.8
Neha	North	130000	110000	5	4.7
Karan	South	70000	90000	2	3.5
Sneha	West	110000	100000	4	4.5
Rohan	North	60000	80000	1	3.2
Pooja	South	125000	100000	5	4.6
Akash	West	80000	90000	3	4.0

BUSINESS ANALYSIS WITH IF

1. Did the salesperson achieve their target?

=IF(C2>=D2,"Achieved","Not Achieved")

2. Give a 10% bonus to salespeople who achieved their target.

=IF(C2>=D2,C2*10%,0)

3. Classify sales performance.

Rules:

Sales >= Target → Achieved

Sales < Target → Not Achieved

Formula:

=IF(C2>=D2,"Achieved","Not Achieved")

BUSINESS ANALYSIS WITH IFS

Classify salespeople:

Sales $\geq$ 140000 → Excellent

Sales $\geq$ 110000 → Good

Sales $\geq$ 90000 → Average

Below 90000 → Poor

Formula:

```
=IFS(  
C2>=140000,"Excellent",  
C2>=110000,"Good",  
C2>=90000,"Average",  
C2<90000,"Poor"  
)
```

BUSINESS ANALYSIS WITH AND

HR wants to identify salespeople who:

- achieved target
- AND have at least 5 years experience

Formula:

```
=IF(AND(C2>=D2,E2>=5),"Eligible","Not Eligible")
```

BUSINESS ANALYSIS WITH OR

Management wants to identify salespeople who either:

- generated sales $\geq$ ₹120,000
- OR have customer rating $\geq$ 4.8

Formula:

```
=IF(OR(C2>=120000,F2>=4.8),"High Performer","Normal Performer")
```

BUSINESS ANALYSIS WITH NOT

Identify employees who have **not achieved** their target.

```
=IF(NOT(C2>=D2),"Needs Improvement","Target Achieved")
```

BUSINESS ANALYSIS WITH IFERROR

Calculate:

Achievement % = Sales / Target

formula:

=C2/D2

Safer formula:

=IFERROR(C2/D2,0)

Then format the result as %.

FINAL CHEAT SHEET

Function	Main Use	Example
SUM()	Total	=SUM(E2:E11)
AVERAGE()	Average	=AVERAGE(E2:E11)
MIN()	Minimum	=MIN(E2:E11)
MAX()	Maximum	=MAX(E2:E11)
COUNT()	Count numbers	=COUNT(E2:E11)
COUNTA()	Count non-empty	=COUNTA(B2:B11)
COUNTBLANK()	Count blanks	=COUNTBLANK(B2:B11)
IF()	One decision	=IF(E2>=50000,"High","Low")
IFS()	Multiple decisions	=IFS(E2>=60,"A",E2>=50,"B")
AND()	All conditions	=AND(E2>=50,F2>=4)
OR()	Any condition	=OR(E2>=60,F2>=5)
NOT()	Reverse condition	=NOT(E2>=50)
IFERROR()	Handle errors	=IFERROR(A2/B2,0)